

SEGMENT ANYTHING MODEL

Zero-Shot Foundation Model for Image Segmentation
CSE 7th Semester | University of Chittagong

Learning Outcomes (OBE)

- ✓ Understand foundation models and zero-shot learning paradigm
- ✓ Explain SAM architecture: image encoder, prompt encoder, mask decoder
- ✓ Apply SAM to real-world use cases (medical, autonomous, robotics, agriculture)
- ✓ Compare zero-shot vs. supervised approaches (SAM vs. nnU-Net)
- ✓ Evaluate segmentation quality: Dice score, sensitivity, specificity
- ✓ Identify fairness issues and ethical considerations in AI models

What is Image Segmentation?

- Partitioning an image into multiple regions (segments)
- Goal: Identify and isolate objects or regions of interest (ROI)
- Applications: Medical imaging, autonomous vehicles, robotics, satellite imagery
- Traditional methods: Thresholding, region growing, watershed, level sets
- Deep learning: FCN, U-Net, DeepLab achieve >85% accuracy on specific tasks
- Challenge: Models are task-specific and domain-specific (poor generalization)

The Problem with Supervised Models

- ✗ Task-specific: Trained only for lung segmentation? Won't work on liver
- ✗ Domain-specific: Trained on MRI? Poor performance on CT scans
- ✗ Data-hungry: Requires 100-1000s of manually labeled images per task
- ✗ Annotation costly: Manual segmentation takes hours per image
- ✗ Time-consuming: Months from data collection → model deployment
- ✗ Poor generalization: Fails on new object types or imaging conditions

Foundation Models: A Paradigm Shift

- ✓ Trained on LARGE + DIVERSE datasets at scale (billions of images/text)
- ✓ Pre-trained ONCE → Reusable for many downstream tasks
- ✓ Zero-shot transfer: Works without task-specific training data
- ✓ Prompt engineering: Guide model behavior with simple inputs (points, text)
- ✓ Examples: CLIP (vision-language), DALL-E (text→image), SAM (segmentation)
- ✓ Economic impact: No need to collect/label data for every new task

Enter Segment Anything Model (SAM)

- Developed by Meta AI Research (2023), published at ICCV 2023
- First true foundation model for segmentation (universal, promptable)
- Trained on SA-1B dataset: 1.1 billion masks from 11 million images
- Can segment ANY object in ANY image with minimal user prompts
- Zero-shot performance competitive with/exceeds supervised models
- Open-source: Apache 2.0 license, free for research + commercial use
- Real-time: ~50ms per prompt on CPU (deployable on edge devices)

SAM Architecture: Three Components

1 IMAGE ENCODER (Vision Transformer ViT-H):

- MAE pre-trained Vision Transformer processes image once
- Outputs 64×64 feature embedding (256 channels)

2 PROMPT ENCODER:

- Encodes user input: points, boxes, masks, text (via CLIP)

3 MASK DECODER (Transformer):

- Combines image + prompt embeddings via cross-attention
- Outputs segmentation mask + confidence scores (~50ms on CPU)

Real-World Use Cases

How SAM Transforms Practice

Use Case 1: Medical Imaging (Lumbar Spine MRI)

PROBLEM:

- Manual segmentation of intervertebral discs (IVD) & vertebral bodies takes 30 mins/image
- Only expert radiologists can do this reliably
- Bottleneck for quantitative imaging research

SAM SOLUTION:

- Radiologist clicks bounding box → SAM segments instantly
- Performance: Dice 0.79 (IVD), 0.88 (VB) with box prompt
- Processing: ~10 seconds per image (vs. 30 minutes manual)
- Reproducible inter-reader agreement (ICC > 0.8)

Use Case 2: Autonomous Vehicles

CHALLENGE: Segment pedestrians, vehicles, roads in real-time

SAM COMPOSITION:

1. Object detector (ViTDet) → outputs bounding boxes
2. SAM + box prompts → precise segmentation masks

ADVANTAGES:

- ✓ Flexible: Swap detectors without retraining SAM
- ✓ Generalizable: Works across cities/weather without new training data
- ✓ Cost-effective: No domain-specific annotation needed for new region
- ✓ Real-time: 50ms per object compatible with autonomous timelines

Use Case 3: Robotics & Object Grasping

PROBLEM: Robot must grasp diverse, unseen objects in real-time

PIPELINE:

1. Eye-gaze tracker → detects where human looks (point)
2. SAM (point prompt) → segments object instantly
3. 3D reconstruction → compute grasp poses (6D gripper orientation)

BENEFITS:

- ✓ Zero-shot: Works on objects never seen before
- ✓ Cross-modal: Same model with RGB, depth, thermal inputs
- ✓ Edge-ready: Real-time on embedded systems (MacBook, Jetson Nano)

Use Case 4: Remote Sensing & Agriculture

CHALLENGE: Segment crop regions across seasons, satellites, crop types

TRADITIONAL: Separate models for wheat/corn/rice, Landsat/Sentinel, dry/wet season

SAM SOLUTION:

- ✓ Single universal model across all crops + all satellites + all seasons
- ✓ Farmers: Click field area → SAM estimates yield, disease coverage
- ✓ Disaster response: Map floods/wildfires without model retraining
- ✓ Scale: Process terabytes of satellite data with one model
- ✓ Impact: Enable precision agriculture in developing countries

Technical Deep Dive

Architecture & Evaluation

Zero-Shot Transfer: 23 Datasets

SAM tested on diverse domains WITHOUT ANY fine-tuning:

MEDICAL: CT, MRI, ultrasound, X-ray, endoscopy

VISION: Object detection, edge detection, instance segmentation

EGOCENTRIC: Hand-object interaction, action segmentation

SATELLITE: Aerial, underwater, paintings, microscopy

RESULTS:

- Single-point Dice: ~ 0.71 (competitive vs. specialized models)
- Human study: SAM masks rated 7-9/10 quality
- Consistency: Similar performance across demographic groups (fairness!)

Comparison: SAM vs. nnU-Net (Lumbar Spine MRI)

METRIC	SAM (Box)	MedSAM	nnU-Net (Supervised)
IVD Dice	0.83	0.79	0.99
VB Dice	0.86	0.88	0.99
Training Data	1.1B mask	Medical	460 images
Training Time	Pre-done	Pre-done	~2 days
Inference Time	~10s	~10s	~1s
Generalization	All tasks	Medical	Task-specific

TRADE-OFF: SAM is 0-shot but lower accuracy; nnU-Net is high accuracy but needs data

Fairness & Bias Analysis

✓ Geographic: Images from 200+ countries (49.8% Europe, 36.2% Asia, 2.8% Africa)

→ Underrepresentation: Africa, Latin America, low-income countries

✓ Gender: Performance on masculine vs. feminine clothing (Dice 0.810 vs. 0.763)

→ Slight bias: Feminine clothing harder to segment

✓ Skin tone: Consistent across Fitzpatrick scale 1-6 (Dice 0.515-0.567)

→ Minimal bias on person segmentation

⚠ LESSON: Foundation models amplify biases in training data!

Better data diversity ↔ Better fairness

Prompt Engineering: Design Choices

POINT PROMPT: Click inside object

Pro: Simple, interactive; Con: Ambiguous (could mean 'person' or 'shirt')

BOX PROMPT: Bounding box around object

Pro: Less ambiguous, +2-3 Dice points vs. points

Con: Requires drawing box (slower than clicking)

MASK PROMPT: Rough mask outline

Pro: Can iterate → Model improves masks

Con: Most effort-intensive

TEXT PROMPT: Natural language (experimental)

Pro: 'segment the lung' (intuitive)

Con: Less reliable than spatial prompts (ongoing research)

Applied Learning

Projects & Implementation

Hands-On Mini Projects

PROJECT 1: Medical Imaging

Download chest X-ray dataset, use SAM+box prompts to segment lungs.

Compare to radiologist annotations. Analyze failure cases.

PROJECT 2: Autonomous Vehicles

Use KITTI dataset, compose YOLOv8 detector + SAM.

Measure latency on edge device (Raspberry Pi). Report speedup.

PROJECT 3: Fairness Audit

Evaluate SAM on MIAP dataset across demographics.

Plot Dice vs. skin tone, age, gender. Identify disparities.

PROJECT 4: Prompt Optimization

Compare point vs. box prompts for 3 different tasks.

Implementation: Python Quickstart

```
from segment_anything import sam_model_registry, SamPredictor
```

```
# Load pre-trained model (once)
```

```
sam = sam_model_registry['vit_h'](checkpoint='sam_vit_h.pth')
```

```
predictor = SamPredictor(sam)
```

```
# Segment from bounding box prompt
```

```
image = cv2.imread('spine_mri.jpg')
```

```
predictor.set_image(image)
```

```
masks = predictor.predict(box=np.array([x1, y1, x2, y2]))
```

```
# Evaluate quality
```

```
dice = (2 * TP) / (2 * TP + FP + FN)
```

```
print(f'Dice: {dice:.3f}')
```

Evaluation Metrics in Detail

DICE SCORE: $2 \times TP / (2 \times TP + FP + FN)$

- Overlap between predicted $\cap$ ground truth masks
- Range: $[0, 1]$, where 1 = perfect overlap
- Acceptable: >0.7 for medical, >0.95 for clinical use

SENSITIVITY (Recall): $TP / (TP + FN)$

- 'Did we find all the positives?' (avoid false negatives)
- Important when: Missing lesions is costly

SPECIFICITY: $TN / (TN + FP)$

- 'Did we avoid false alarms?' (avoid over-segmentation)
- Important when: False alarms lead to unnecessary procedures

Limitations & Future Work

CURRENT LIMITATIONS:

- ✗ Fine structures: May miss small details (thin blood vessels, hair)
- ✗ Boundary quality: Not as crisp as specialized models
- ✗ 2D-only: Must apply slice-by-slice for 3D volumes
- ✗ Requires prompts: Cannot fully automate (though this is flexible!)

EMERGING SOLUTIONS:

- ✓ SegmentAnyBone: Fine-tuned on bone anatomy (Dice >0.85)
- ✓ MA-SAM: Modality-agnostic (works on CT, MRI, ultrasound)
- ✓ 3D SAM: Process entire volumes at once (research stage)

Synthesis

What You Should Take Away

Key Takeaways

1. Scale + Diversity = Generalization

1.1B masks from diverse images enable zero-shot transfer

2. Paradigm Shift: From 'Train for Task' to 'Compose for Task'

Foundation models reduce time from months to minutes

3. Prompting is Human-Aligned

Points/boxes are more intuitive than tuning hyperparameters

4. Fairness Requires Active Monitoring

Check bias across demographics; don't assume 'one model for all'

5. Composability Enables Rapid Development

Combine SAM with detectors, classifiers, 3D reconstruction → new capabilities

Questions?

What excites you about SAM?